

ROPE HAS A SPECTRAL BUDGET

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ABSTRACT

The number of rotary channels does not uniquely determine the positional responses available to an attention head. We study whether frequency allocation can reduce the cost of a smaller rotary budget while keeping total query/key width, model size, and training resources fixed. A finite-distance analysis bounds the spectral tail contributed by nearly repeated slow-frequency subspaces. Under a bounded coefficient norm, this limits responses outside their shared span; a controlled softmax result states the additional conditions needed to turn that restriction into distributional error. These are conditional basis results, not language-model loss bounds. We instantiate the question with a fixed-support anchored EVQ-Cosh allocation and a six-arm factorial design: 32 versus 16 rotary pairs, geometric versus analytic allocation, and two 16-pair frequency-subset controls. Existing fixed-support training and single-budget MLA results motivate the experiment but do not identify a budget interaction. The new paired training study is in progress. Whether analytic allocation preserves long-context quality at half the rotary width, without impairing the training window or losing to simple subset selection, remains an empirical question.

1 FREQUENCY ALLOCATION UNDER A FIXED ROTARY BUDGET

An attention head can devote fewer dimensions to rotation without removing its content channels. Does the way those remaining frequencies are placed determine how much long-context quality is lost? We study this question at fixed total query/key width, head count, parameter count, and training budget. The resource is the number K of rotating two-dimensional pairs; the remaining coordinates retain identity rotation. This is a representation-efficiency question. In ordinary multi-head attention it does not, by itself, reduce the KV tensor.

The relevant comparison is stronger than showing that a new frequency table improves a geometric table at one width. A budget-specific claim requires an allocation-by-budget interaction. A practical half-budget claim additionally requires comparison with simple strategies that retain selected frequencies from the full table. We therefore retain both a high-frequency subset and a uniform subset as main controls. They may explain the entire benefit of reducing rotation, in which case a special advantage for EVQ would be unsupported.

Slow rotary pairs offer a useful starting point for theory. With content coefficients fixed, their positional response subspaces approach the same span of a constant and a linear function of distance. We quantify the residual energy outside that span on a finite distance distribution, bound the spectral tail of the complete table, and impose a coefficient budget before discussing approximation error. A softmax bridge then applies to a controlled read with bounded centered logits. Content-dependent coefficients and communication through other layers remain outside that argument; only model experiments can determine whether the restriction is consequential in language modeling.

Our sole analytic candidate is anchored EVQ-COSH, with one fixed shape strength across budgets. Its previous surrogate derivation supplies a reproducible construction; it does not establish optimality for the new budget objective. Nor does an asymptotic quantization bound imply increasing gains as K shrinks: when $K = 2$ and both endpoints are fixed, EVQ and geometric tables coincide.

Evidence and open test. A prior three-seed fixed-support intervention establishes that interior allocation can change learning outcomes (App. C.1). A separate three-seed MLA study provides positive long-range evidence at a single small rotary budget (§5). Neither establishes what happens

when only rotary width changes. The new six-arm experiment isolates that question, with same-target long-document NLL, training-window retention, and a position-preserving remote-prefix control. Its results are pending; the current manuscript presents the theory and registered experiment without claiming half-budget preservation.

2 FINITE-DISTANCE CONSTRAINTS ON A ROTARY BUDGET

Fixed width and pair identities. For total head width d_h , retain K rotating pairs and $d_h - 2K$ unrotated coordinates. The attention logit is

$$s_{ij} = \frac{1}{\sqrt{d_h}} \left[\sum_{k=1}^K (q_i^{(k)})^\top R_{\omega_k} (i-j) k_j^{(k)} + (q_i^N)^\top k_j^N \right]. \quad (1)$$

All parameter shapes and the temperature remain fixed. We parameterize the active frequencies by

$$\omega_k = \exp[-(a + Rz_k)], \quad z_0 = 0, \quad z_{K-1} = 1. \quad (2)$$

Same- K comparisons at fixed (a, R) isolate allocation. Cross- K comparisons change the division between rotating and nonrotating content coordinates.

Stable finite-distance geometry. Let $t = \Delta/L$ on $\Delta = 0, \dots, L-1$ with a declared positive distribution μ . The main geometric diagnostic uses causal-pair weights $\mu(\Delta) \propto L - \Delta$ and also reports uniform weights. These weights are not learned attention probabilities. For $x = \omega L$, let $v_x = [\cos(xt), \sin(xt)/x]$, with $v_0 = [1, t]$. Weighted QR yields an orthonormal basis Q_x for each pair’s subspace. Set $A_\Omega = [Q_{x_1}, \dots, Q_{x_K}]$ and $\Gamma_\Omega = A_\Omega^* A_\Omega$, so $\text{tr } \Gamma_\Omega = 2K$. Let P_0 project onto $U_0 = \text{span}\{1, t\}$ and define $e(x) = \|(I - P_0)Q_x\|_{\text{HS}}^2$.

Proposition 1 (Finite budget and bounded-coefficient response). *Suppose $m \geq 1$ active pairs satisfy $x_k \leq \varepsilon$. Let $\eta = \sum_{k: x_k \leq \varepsilon} e(x_k)$ and $r_0 = 2(K - m + 1)$. For the eigenvalues $\lambda_1 \geq \dots \geq \lambda_{2K}$ of Γ_Ω ,*

$$\sum_{r > r_0} \lambda_r \leq \eta. \quad (3)$$

Let U contain U_0 and all non-slow pair subspaces. Then, for every $\|c\|_2 \leq B$ and target response b ,

$$\|(I - P_U)A_\Omega c\|_\mu \leq B\sqrt{\eta}, \quad \inf_{\|c\|_2 \leq B} \|b - A_\Omega c\|_\mu^2 \geq (\|(I - P_U)b\|_\mu - B\sqrt{\eta})_+^2. \quad (4)$$

Replace each slow block with $P_0 Q_x$. The resulting matrix has rank at most r_0 and squared approximation error η , proving Eq. (3). Projection, the operator-norm bound by the Hilbert–Schmidt norm, and the triangle inequality give Eq. (4). When no pair is slow, the tail statement is vacuous with $r_0 = 2K$ and $\eta = 0$.

What the constraint means. Several slow pairs occupy distinct coordinates while adding little energy outside a shared positional span. This does not make their content coefficients redundant. The bound becomes informative only if the target needs a direction outside U and the available coefficient norm cannot compensate. In the normalized basis, raw coefficients $C \cos(xt) + D \sin(xt)$ become $c = R_x[C, xD]^\top$ when $v_x = Q_x R_x$. The factor x is essential; the bound cannot simply assume the same B from a shared Q/K norm across trained models.

From logits to a controlled read. For $p_f(d) \propto \mu(d)e^{f(d)}$, centered target and model logits bounded in magnitude by M satisfy

$$D_{\text{KL}}(p_b \| p_f) \geq \frac{e^{-2M}}{2} \|b - f\|_\mu^2. \quad (5)$$

Together with Eq. (4), this gives a conditional softmax fitting obstruction (App. A). Ordinary single-query attention uses uniform μ here; no $\log \mu$ bias is added to the Transformer. This theorem concerns a controlled position response, not the NLL of a deep model with varying content and alternative information paths.

Falsifiable implication. Reallocation can help only when the original basis lacks relevant directions and the trained model can exploit the new ones without larger content or optimization costs. A higher static rank alone is insufficient. The experiment therefore tests whether the budget loss is smaller under EVQ, $(\ell_{G16} - \ell_{G32}) - (\ell_{E16} - \ell_{E32}) > 0$, without predicting monotonic gains over all K or guaranteeing lossless halving.

3 ONE FIXED ANALYTIC ALLOCATION

Set $d_h = 64$, $K_{\text{ref}} = 32$, $a = 0$, and $R = (31/32) \log(500000)$. For either $K = 16$ or 32 , the geometric grid is $z_k^G = k/(K-1)$. The anchored EVQ grid is

$$u_k = \frac{k + 1/2}{K}, \quad q_k = 1 - \frac{\text{asinh}[(1 - u_k) \sinh \tau]}{\tau}, \quad z_k^E = \frac{q_k - q_0}{q_{K-1} - q_0}, \quad \tau = \sqrt{2}. \quad (6)$$

The same strength is used at both budgets. It follows the existing full-head reference convention $64/\sqrt{2048}$, not a fitted strength or an optimum implied by Proposition 1. Previous variational optimality remains limited to its original surrogate (App. D).

Every arm retains 32 frequency slots and split-half pairs $(j, j + 32)$. The first K slots use the active table; other slots have frequency zero. The full-head kernel and temperature $1/8$ are unchanged. P16 retains the fastest 16 G32 frequencies; U16 retains indices $0, 2, \dots, 30$. These subset controls do not share the slow endpoint of the anchored G/E arms and therefore do not enter the pure-allocation estimand. They remain necessary practical baselines for whether the analytic construction adds value.

4 PREPARED BUDGET EXPERIMENT

Status. The implementation and runtime probe are prepared; formal training is not currently scheduled while the central modern-attention question is being tested. Table 1 is a design table, not a results table. No newly trained budget-arm score is available in this revision.

Arm	Rotary pairs	Allocation
G32	32	Fixed-support geometric
E32	32	Same-support anchored EVQ
G16	16	Same-support resampled geometric
E16	16	Same-support anchored EVQ
P16	16	Fastest half of G32
U16	16	Every second G32 frequency

Table 1: Prepared six-arm comparison. All arms retain 12 heads of total width 64 and identical parameter shapes. Seeds 42, 137, and 256 would be paired if this comparison is activated. The six-arm queue is not the current research priority. No entries denote measured model quality.

Training. We reuse the 12-layer, width-768 decoder with MLP width 3072 and vocabulary 50304. The default is 7629 updates of 65536 input tokens each, or 499,974,144 tokens per arm, at training length 2048. Initialization, token prefix, row order, optimizer, learning-rate schedule, and update count are paired within each seed. The optimizer is fused AdamW with $\beta = (0.9, 0.95)$, weight decay 0.01, peak/minimum rates $6 \times 10^{-4}/6 \times 10^{-5}$, 762 warmup updates, and clipping 1.0. The final common budget is frozen from actual throughput and available resources before observing method comparisons. A shortened, unstable training run is reported as a pilot rather than a converged budget frontier.

Same-target evaluation. Before comparison, freeze 512 independent long documents from the held-out FineWeb-Edu shard. Each has at least 8193 tokens, permitting 8192 inputs and one shifted final label. Contexts of 512, 2048, 4096, and 8192 tokens predict the same final 256 targets. The primary metric is document-mean NLL at 4096; PPL is its exponential. Report 2048 same-target and whole-window NLL for retention, and 8192/512 as trends. Every evaluation uses the training table and unit gain, without test-time scaling or routing.

Remote-context control. For the 4096-token input, replace the first 3584 tokens with a fixed independent document’s prefix while retaining the last 512 input tokens, labels, length, and position IDs. The same document derangement applies to all arms. Define $C_A = \ell_A^{\text{replaced}} - \ell_A^{\text{intact}}$ and $I_{\text{context}} = C_{E16} - C_{G16}$. Greater sensitivity supports useful remote context only alongside lower intact NLL; sensitivity to a wrong prefix alone is not evidence of better use.

Contrasts and decision. At the primary length, report per-seed

$$D_{16} = \ell_{G16} - \ell_{E16}, \quad D_{32} = \ell_{G32} - \ell_{E32}, \quad J = D_{16} - D_{32}, \quad (7)$$

$$H = \ell_{E16} - \ell_{G32}, \quad P = \ell_{P16} - \ell_{E16}, \quad U = \ell_{U16} - \ell_{E16}. \quad (8)$$

The practical and noninferiority margins are 0.02 nat/token. Half-budget preservation requires the interval for H below the margin, native-window retention, and comparison with both subset baselines. A positive J separately supports mitigation of budget loss. If G16 does not lose quality against G32, there is no observed bottleneck to recover. A recovery ratio is reported only when that denominator is stable and at least 0.02.

Three training seeds are the replication units. Report all seed contrasts, their mean and standard deviation, and any Student- t interval with the normality assumption and two degrees of freedom explicit. Document bootstraps condition on the trained checkpoints; documents are not training replications. The main plot will show both G/E budget points, individual seeds, and P16/U16, with native-window quality in a separate panel. Pending runs are not plotted as predicted scores.

5 EXISTING EVIDENCE AND ITS SCOPE

Prior fixed-support 151.9M training supports an allocation effect but used length 256 and does not supply a baseline for the new 2K experiment (App. C.1). The older 50.9M factorial changes head count, head dimension, and shape strength; it does not isolate the present budget intervention.

The existing 432M MLA-style study has $d_{\text{rope}} = 32$, hence $K = 16$. Across three seeds, raw 16K PPL is 138.8 for geometric RoPE and 95.6 for EVQ; 8K is 35.4 versus 35.8. Under its shared run-specific wavelength blend, 16K is 117.9 versus 71.1. This provides a relevant single-budget result and a reason to test native retention, but no allocation-by-budget interaction. Existing 750M training and retrieval results provide separate scale evidence. Their original protocols and evidence boundaries remain in the supporting appendices; neither is relabeled as a replication of the new experiment.

6 RELATED WORK

Partial rotation and frequency selection. Slow RoPE bands can support content matching rather than distinct position responses (Barbero et al., 2025). MHA2MLA uses partial rotation and frequency selection as part of converting pretrained attention to a compressed latent architecture (Ji et al., 2025). Thus neither reducing rotary dimensions nor the importance of selection at a small width is a new observation here. Our proposed comparison holds the complete attention architecture fixed and asks whether an anchored analytic allocation improves on explicit high-frequency and uniform-subset baselines under the same training budget.

Learned and data-dependent frequency use. LeRoPE learns frequencies and compares them with geometric and partial RoPE (Karypis et al., 2026); frequency usage also depends on the data’s dependency structure (Wu et al., 2026). These results rule out treating a static spectral diagnostic as a universal utility metric. Our candidate uses one fixed analytic shape without a frequency search, but that distinction alone does not establish a practical advantage. The six-arm experiment must determine it.

Context extension. Position interpolation, YaRN, LongRoPE, and MrRoPE modify the treatment of positions or frequencies during extension (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Tian et al., 2026). MrRoPE studies mixed-radix transports of an existing table; the present experiment studies training-time allocation under changing rotary width. These mathematical design

spaces can overlap. MrRoPE is relevant extension prior work rather than a substitute for the budget controls here. Any future zero-training superiority claim would require its own matched strong-baseline comparison.

7 SCOPE AND LIMITATIONS

The new claim to test is whether allocation reduces the long-context cost of a smaller rotary budget under exact architecture and training controls. The finite-distance results establish conditional restrictions on position responses. They neither rank all trained models nor guarantee that EVQ beats frequency-subset selection. A negative result against P16 or U16 must remain in the main comparison and change the interpretation of the construction.

A successful experiment would establish representation efficiency for the specified decoder and evaluation range. It would not establish universal scaling with rotary width, superiority in zero-training extension, or an automatic halving of KV memory. Decoupled rotary keys in MLA may connect width to cache cost, but the present full-width MHA experiment does not measure that systems benefit. The separate KLD follow-up studies recurrent memory and does not form a second method or supporting result for this paper.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

The new six-arm experiment is specified by a frozen protocol, full frequency tables, and same-target input construction. Implementation checks cover fixed pair identities, inactive coordinates, parameter initialization, temperature, causal label alignment, and remote-prefix replacement. Training records must include data and frequency hashes, paired row order, optimizer state, token budget, and resumable progress. The primary analysis reports training-seed and conditional document uncertainty separately. At this revision, CPU checks and the implementation are available; new model results remain pending. Previous experiments retain their own protocols and artifact-availability limitations in the supporting appendices. In particular, the single-budget MLA aggregate does not imply availability of its original training checkpoints.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

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A FINITE-BUDGET BOUNDS: CONDITIONS AND PROOFS

T1: a finite slow-pair bound. Let G_0 be the Gram matrix of $[1, t]$ in $L^2(\mu)$, $\sigma_0 = \sqrt{\lambda_{\min}(G_0)} > 0$, and $h_\mu^2 = \mathbb{E}_\mu[t^4/4 + t^6/36]$. If $0 < x \leq \varepsilon$ and $\varepsilon^2 h_\mu < \sigma_0$, then

$$e(x) \leq \frac{x^4 h_\mu^2}{(\sigma_0 - \varepsilon^2 h_\mu)^2}. \quad (9)$$

Indeed, $|\cos(xt) - 1| \leq x^2 t^2/2$ and $|\sin(xt)/x - t| \leq x^2 t^3/6$ give $\|v_x - v_0\|_{\text{HS}} \leq x^2 h_\mu$. Singular-value perturbation gives $\sigma_{\min}(v_x) \geq \sigma_0 - x^2 h_\mu$. Since $(I - P_0)v_0 = 0$, orthonormalization bounds the residual by $x^2 h_\mu/(\sigma_0 - x^2 h_\mu)$. Squaring and bounding the denominator proves the result. Exact weighted-QR residuals can be substantially sharper.

T2 and T3. The rank approximation in Proposition 1 has residual squared norm η . The best rank- r_0 approximation cannot be worse, and the squared singular values of A_Ω are the eigenvalues of its Gram matrix. Moreover $(I - P_U)A_\Omega$ has operator norm at most $\sqrt{\eta}$. For any bounded c , project $b - A_\Omega c$ onto $U^\perp$ and apply the reverse triangle inequality. This yields the positive-part lower bound. For a declared threshold $\zeta > 0$, at most $r_0 + \eta/\zeta$ eigenvalues can exceed ζ ; this diagnostic depends on the chosen tolerance and is not model accuracy.

T4: a nonlocal softmax bound. Assume μ is a probability measure with positive support. Center b and f under μ , with both bounded by M in absolute value. Along $g_s = (1 - s)b + sf$, the tilted probability satisfies $p_{g_s} \geq e^{-2M}\mu$. For $\delta = f - b$,

$$\text{Var}_{p_{g_s}}(\delta) = \inf_c \mathbb{E}_{p_{g_s}}(\delta - c)^2 \geq e^{-2M} \inf_c \mathbb{E}_\mu(\delta - c)^2 = e^{-2M} \|\delta\|_\mu^2. \quad (10)$$

The Hessian of log-partition is this variance. Integrating its quadratic form along the segment with weight $1 - s$ gives $D_{\text{KL}}(p_b \| p_f) \geq e^{-2M} \|b - f\|_\mu^2/2$. Because U includes constants, centering leaves the projected residual unchanged. If the uncentered model response has coefficient norm at most B , T3 consequently gives

$$D_{\text{KL}}(p_b \| p_f) \geq \frac{e^{-2M}}{2} (\|(I - P_U)b\|_\mu - B\sqrt{\eta})_+^2. \quad (11)$$

This is a bound for a specified controlled read. Independent values and a query that cannot access them are needed for a value-reading interpretation; it is not a lower bound on a Transformer language-model loss.

B PREVIOUSLY REPORTED ALLOCATION EVIDENCE

The following supporting material retains the original experiments and theoretical objects. It supplies no observations from the new six-arm budget experiment.

C SUPPORT-ALLOCATION CAUSAL DECOMPOSITION

C.1 FIXED SUPPORT ISOLATES STRUCTURED ALLOCATION

A clean interior-allocation intervention. Within each of three 151.9M training seeds, both arms share architecture, initialisation, token order, optimiser, LR schedule, global batch, a 499,974,144-token budget, and 32 evaluation anchors. The baseline is FMROPE at fixed training support, implementing Oka et al. (2026, §6.1) with $\theta = L_{\text{train}} = 256$ and geometric exponents. Anchored EVQ-COSH ($\tau=4$) normalises its quantiles to the same extrema and log-span, so only the $K-2 = 30$ interior frequencies move while retaining the deployed Cosh table’s z . Every training seed favours anchored EVQ-COSH at every OOD length. Anchored EVQ-COSH minus FMROPE NLL is $+0.026$ at the 256-token training length and $-0.281/-0.176/-0.146$ at 512/1K/2K (Fig. 1b). No geometric base can reproduce this intervention once the sampled support is pinned: *fixed support + different $z \Rightarrow$ different trained-model behaviour*. Thus z is a separately identifiable coordinate, not a disguised base change.

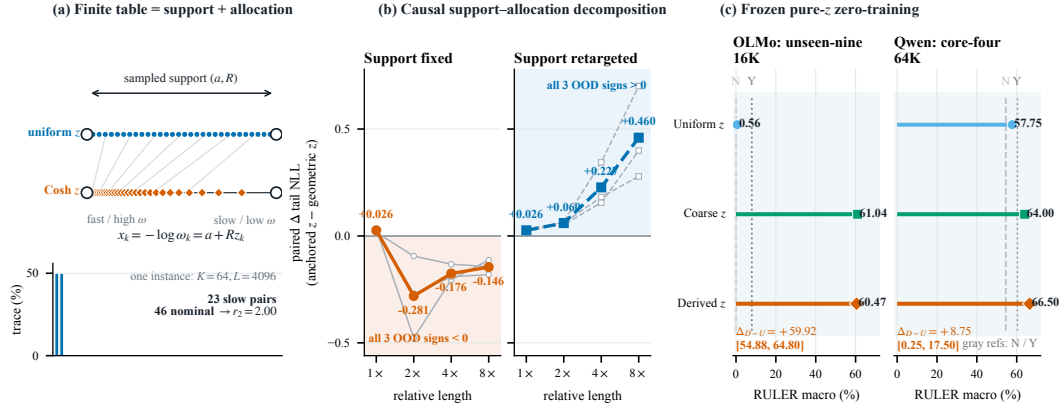


Figure 1: Previously reported allocation evidence, reproduced for reference. The fixed-support training and frozen-table panels do not intervene on rotary pair count and therefore cannot establish the new budget interaction.

Target-aware support identifies the second coordinate. The second condition permits the declared L_{target} to retarget each arm’s sampled support while preserving its normalised interior shape, giving the paper-faithful FMRoPE control its intended target-aware input. The ordering then reverses in all three seeds: anchored EVQ-COSH minus FMRoPE NLL is $+0.060/+0.227/+0.460$ at 512/1K/2K. The two conditions answer different causal questions. Fixed support isolates the effect of structured interior allocation; target-aware retargeting shows how support selection changes which allocation is effective. Their reversal establishes support and allocation as distinct but interacting coordinates of the same finite table. The full protocol, per-seed contrasts, and both support conditions are in App. H.

Configuration and shape breadth. The fixed-support result is not confined to one base, head width, or Cosh strength. A pre-specified 50.9M factorial crosses two bases, two training lengths, three head dimensions, and three paired seeds while pinning sampled endpoints and log-span in every arm. Three Cosh strengths and a deformation-matched exponential provide four structured non-geometric curves. The strongest Cosh arm improves the weighted OOD objective in 10/12 configurations; the matched exponential does so in 9/12 and closely tracks the formula-Cosh arm. This broadens the finding from one curve to a family of structured allocations: Cosh supplies the analytic construction, while the 151.9M intervention above remains the effect-size owner. The complete matrix is in App. Table 12; model-scale persistence is tested separately in §E.4.

D THEORY

Finite RoPE design is a constrained basis-allocation problem. With pair count and sampled support fixed, a table chooses where to place phase resolution. We first account for that finite budget exactly, then locate the slow-end redundancy of geometric sampling and derive EVQ-COSH as a closed-form reallocation. Finally, we connect the designed basis to the stage at which weights exploit it; capability is established by the controlled interventions, not inferred from static geometry.

D.1 A FINITE SPECTRAL BUDGET IS A DESIGN PROBLEM

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}} = d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

We reserve z for the support-normalised coordinate in Eq. (2). Normalising a sampled geometric table gives $z_k = k/(K-1)$, so fixing (a, R) fixes the geometric family and a non-uniform interior allocation is a distinct fixed-support design. Standard implementations use raw exponents $u_k = k/K$, deployed EVQ-COSH uses midpoint quantiles $u_k = (k+1/2)/K$, and endpoint-inclusive diagnostic grids are labelled explicitly.

A rotary pair contributes $f_\omega(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (12)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_ω and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ whenever $S_\omega, S_\nu \succ 0$, the singular values of $Q_{\omega\nu}$ are the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (13)$$

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. F).

Block-whitening every pair gives a global correlation Gram Γ with diagonal blocks I_2 . Write $\bar{c} = [K(K-1)]^{-1} \sum_{i \neq j} c_{ij}$.

Theorem 2 (Spectral budget identity). *$\text{tr} \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 effective rank of the positional basis is exactly*

$$r_2(\Gamma) = \frac{(\text{tr} \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (14)$$

Practically, this identity exposes the design budget. For fixed K , sampled support (a, R) , and separation measure μ , substituting $\omega_k(z) = \exp[-(a + Rz_k)]$ into the block-whitened Gram $\Gamma_\mu(z)$ changes how the same trace $2K$ is distributed across positional directions. Block whitening removes per-channel energy and conditioning: the theorem characterises the basis’s Rényi-2 effective dimension, while trained-model protocols establish what that basis enables.

D.2 SLOW-END REDUNDANCY GIVES A DESIGN DIRECTION

Proposition 3 (Low-frequency collapse). *As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$. For slow channels $x=\omega L$, $y=\nu L$, and $\epsilon = \max\{|x|, |y|\} \rightarrow 0$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric for a fixed p with nondegenerate support on at least three distances, $F = \text{diag}(p) - pp^\top$ annihilates the constant direction, and the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.*

Practically, geometric slow-end sampling provides diminishing new positional directions. This motivates thinning that region while preserving support and multiscale coverage; the channels may still carry content, and the static rank is not a behavioural selector. As a representative severe-regime instance, at $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 standard-grid pairs with $\omega L \leq 1$ occupy 46 nominal dimensions but have block-whitened $r_2=2.00$ under the uniform separation prior. The full-table identity also applies outside this asymptotic regime; additional views appear in App. Fig. 4.

D.3 A CLOSED-FORM REALLOCATION

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a finite allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. The exact canonical-correlation objective is oscillatory, so we use the convex modelling surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (15)$$

with $\alpha > 0, \beta \geq 0$. The first term limits channel-load concentration; the Green kernel $\min(\phi, \psi)$ penalises pairs that are both slow, following the direction of Proposition 3.

Theorem 4 (Closed-form optimum under the surrogate). *Under (15) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (16)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Practically, the theorem turns the slow-end design direction into an analytic allocation rather than a search problem. The following inverse CDF instantiates that continuous solution as a finite table.

Corollary 5 (Closed-form quantiles). $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (17)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

For $\tau > 0$, ρ_τ decreases across log-frequency and crosses the uniform density once (Lemma 8); the quantiles therefore place more samples on the fast side and thin the redundant slow side. This gives EVQ-COSH explicitly, without learned positional parameters or numerical search. Its uniqueness is conditional on (15); the operating rule, finite multiplier evidence, grid conventions, and exact-range endpoint embedding are in Apps. F.9 and H. The cosine-feature diagnostic is reported separately in App. F.6; trained interventions establish behaviour.

D.4 A DESIGNED BASIS MUST BE EVALUATED AT ITS INSTALLATION STAGE

A designed table fixes the positional basis; model capability depends on the coefficients that exploit it. Controlled fixed-support protocols show that designed reallocations can strongly improve capability in the tested protocols. The training intervention in §C.1 evaluates anchored EVQ-COSH, while the fully frozen study in §E.1 evaluates separate derived and coarse profiles; they supply likelihood and capability evidence, respectively.

The installation stage is visible in a frozen 50M weights-by-table crossing (Table 2). The self-consistent Geo and EVQ-COSH systems have PPL 7.14 and 7.16, while both mismatched cells degrade. In particular, installing EVQ-COSH under Geo-trained weights raises block-whitened r_2 from 4.57 to 12.54 but worsens PPL from 7.14 to 76.20; the loss interaction is -3.537 (bootstrap CI $[-5.17, -3.04]$).

Table 2: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Thus static geometry diagnoses the designed basis but does not rank trained systems: weights learn coefficients in the basis installed during training or adaptation.

Theorem 6 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta) B = \mathcal{R}_\Omega(\Delta)$ for all Δ in an open interval containing zero, then Ω' and Ω have the same frequency multiset, up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Practically, a nontrivial new table cannot be made exactly function-equivalent by fixed, position-independent invertible Q/K maps: the design must be evaluated with the weights, adapter, or operator that will use it. At $\Delta=0$ the hypothesis gives $A^\top B = I$; differentiating the resulting similarity at zero preserves the generator spectrum $\{\pm i\omega_k\}$ (App. F). The theorem does not exclude approximate finite-data adaptation or broader paths, which are tested in §E.3.

Together, the budget identity defines what can be reallocated, the collapse law locates redundant slow-end sampling, and the variational theorem supplies a closed-form table before its coefficients are learned. The fixed-support and lifecycle experiments test designed allocations with weights fully frozen, under matched adaptation, and through from-training co-adaptation (§E).

E THE ALLOCATION COORDINATE ACROSS MODEL-BUILDING REGIMES

We intervene on the allocation rule at three stages of the model lifecycle. With a mature checkpoint fully frozen, only the installed allocation changes; under matched low-rank adaptation and from training, the resulting weight changes are responses to that allocation rather than separate method variants. We keep likelihood, capability, and uncertainty units separate throughout.

E.1 FROZEN MATURE MODELS: PURE- z ZERO-TRAINING INTERVENTION

Allocation alone changes a mature model. At matched support, amplitude, checkpoint, rows, decoder, precision, and hardware, changing only interior z raises OLMo nine-task RULER at 16K from 0.56% to 60.47%; a coarse label-free allocation reaches 61.04%. On Qwen at 64K, the corresponding fixed-support change raises core-four RULER from 57.75% to 66.50% (Fig. 2). The intervals for derived minus uniform allocation are [54.88, 64.80] and [0.25, 17.50] points, respectively. These uniform, coarse, and derived mature tables are three values of the same fixed-support coordinate; they are not EVQ-COSH tables, and the coarse-versus-derived comparison does not identify a unique profile (App. J.2).

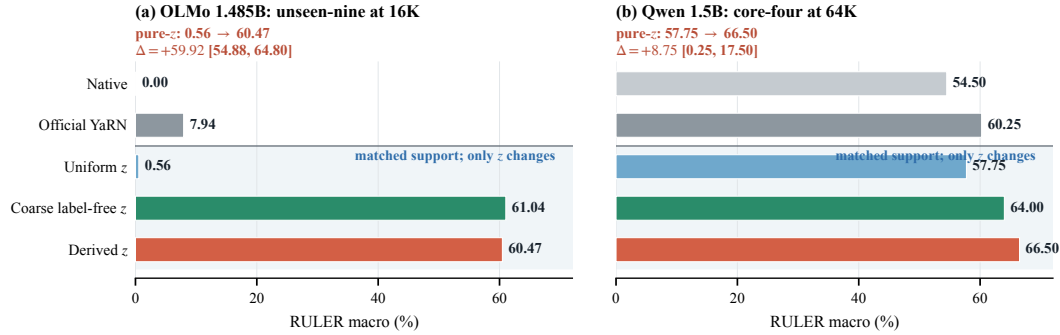


Figure 2: **Changing only interior allocation can transform a frozen checkpoint.** Grey reference rows retain their own support or transport; the shaded block holds support, amplitude, checkpoint, rows, decoder, precision, and hardware fixed while changing only z . Derived minus uniform allocation is +59.92 points ([54.88, 64.80]) on OLMo and +8.75 points ([0.25, 17.50]) on Qwen. Each task has 20 rows; intervals condition on each fixed checkpoint and task set through evaluation-row resampling. Exact values and the coarse-control contrasts are in App. J.2.

For implementation, given Native frequencies ω_k , extension factor s , and fixed displacement weights $m_k \in [0, 1]$, the mature intervention installs

$$\hat{\omega}_k = \omega_k [(1 - m_k) + m_k/s], \quad \hat{z}_k = \frac{-\log \hat{\omega}_k - \hat{a}}{\hat{R}}, \quad (18)$$

with the fast and slow endpoints pinned. Equation (18) defines the pure- z block above; the complete deployment policy below is a separate bundled intervention.

E.2 ZERO-TRAINING DOWNSTREAM CAPABILITY AND NATIVE-WINDOW RETENTION

Building on the pure- z result, the zero-training system bundles one frozen factor-four long table, a fixed long attention amplitude, and a session-static Native/long route chosen before prefill. It adds no learned parameters or training tokens, but its combined gain must not be assigned to allocation alone. Table 3 separates its likelihood, capability, downstream, and routing endpoints; none of these metrics is pooled.

The route retains the exact Native path when a request fits the model window; otherwise it uses one deployment-frozen factor-four profile for the session. This practical policy is neither the pure- z contrast in Fig. 2 nor the EVQ-COSH construction used in the adapted and from-training studies. Full task-level results, row identities, and the allocation-versus-routing controls are in App. J.2.

Table 3: **Zero-training system endpoints.** The bundled policy chooses one Native or frozen long route before prefill and retains it for the session. Rows use different metrics and experimental units and are not pooled. The policy is distinct from the pure- z block in Fig. 2 and from EVQ-COSH.

Endpoint	Reference	Bundled policy	Evidence role
FineWeb-Edu final-1,024 tail NLL, 4K ($n=128$)	Native 2.7538; static long 2.8774	2.7538 (Native route)	in-window routing
PG-19 tail NLL, 8/16K	official YaRN 3.4404/3.7955	3.1060/3.0977	natural likelihood
RULER-13 macro, 8/16K	official YaRN 23.82/5.88%	67.72/54.40%	task capability
Qasper (Dasigi et al., 2021) / 2Wiki token-F1, full 200	official YaRN 18.03/25.69%	24.57/26.66%	natural downstream

E.3 MATCHED ADAPTATION: LENGTH TRANSFER AND CAUSAL SOURCE USE

Matched adaptation asks whether pretrained representations can learn to use an EVQ-COSH allocation. Starting from protocol-matched Native and EVQ-COSH Stage-A parents, the 1.485B selective-Q/K arms freeze inherited V/O LoRA and continue only Q/K for 300 steps with identical data, order, optimiser, trainable count, and phase exposure. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH token-F1 is 0.07/21.48% and 0/8.57% at 8/16K, versus 25.99/24.84% at 4K (Ho et al., 2020; Bai et al., 2024). An independent 13-family RULER continuation gives the same task-family crossover: 2.02/31.63% and 0.38/5.03% at 8/16K, versus 72.19/42.44% at 4K (Hsieh et al., 2024). Training and evaluation rows are disjoint within each task family (App. J).

At 8B, a single matched rank-64 Q/K/V/O LoRA pair provides a complementary causal endpoint. At true 16K, the EVQ-COSH arm raises median target-block hit@16 from 18.75% to 64.06%; position-preserving removal of the remote gold block changes answer NLL by -0.0095 / $+1.5055$ (Fig. 3). Thus the adapted EVQ-COSH model uses the remote source in these cases; this routing/probability result is distinct from autoregressive task accuracy.

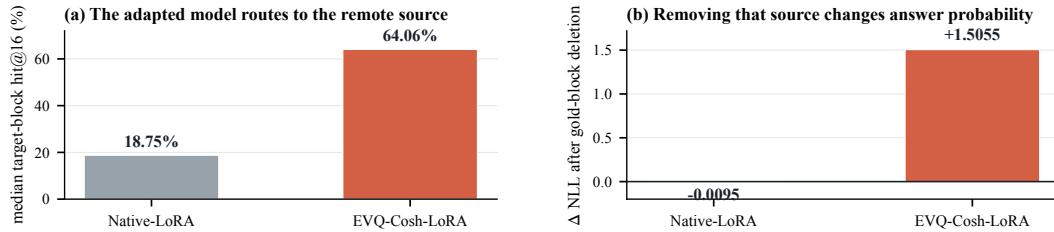


Figure 3: **Matched adaptation converts the new basis into causal remote-source use.** In one matched trained pair, ten frozen true-16K cases measure (a) median target-block hit@16 over 32 retrieval heads and (b) the answer-NLL change after deleting decode attention to the remote gold block from every head while preserving prompt tokens, rotary KV indices, and block geometry. These are routing and teacher-forced probability endpoints, not training replications or autoregressive accuracy (App. J.3).

E.4 FROM TRAINING: CO-ADAPTATION, SYSTEMS, AND IN-WINDOW BEHAVIOUR

The paired three-seed intervention in §C.1 owns the fixed-support causal claim. The studies in Table 4 instead test whether an installed allocation rule remains consequential under scarce rotary channels, continued or from-initialisation full-parameter training, and bidirectional video. The fixed-support studies own causal attribution; these rows test stage-specific persistence, with each protocol’s execution boundary retained in the table and appendix.

The text protocols exhibit a recurring in-window/long-range crossover, while the video-DiT comparison improves both training and extrapolated frames. These rows therefore establish architecture-,

Table 4: **From-training and system breadth.** Geo/EVQ-COSH entries are reported within each protocol; ΔNLL is EVQ-COSH minus Geo. Rows have different experimental units and endpoints and are not pooled. The allocation rule is the intended scientific intervention; causal attribution is owned by the fixed-support studies, while these rows establish stage-specific persistence.

Protocol (unit)	Original window	Longer positions / capability	Maximum role
432M MLA, PPL $35.4/35.8$ at PPL $138.8/95.6$ at 16K			scarce-channel system
$K=16$ (3 seeds) 8K			persistence / capability
750M continuation PPL $22.0/22.3$ at PPL $45.1/24.4$ at 16K; AR exact $0/77.5\%$ (1 pair, seed 42) 4K		at 8K	pretraining-scale persistence
1.485B from initial- $\Delta\text{NLL} + 0.0724/ + \Delta\text{NLL} - 0.0437/ - 0.1351$ at 8/16K isation (1 same-init 0.0381 at 2/4K comparison)			
129.6M video DiT train-frame MSE all/far extrapolated MSE $-16\%/ - 35\%$ (seed 42) -21%			cross-modal breadth

scale-, and modality-specific persistence, not a pooled effect or scaling law. Complete PPL trajectories, strict generation endpoints, and per-seed MLA values are in Apps. G, J, and K.1.

F PROOF DETAILS

F.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (19)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. The metric is consequently attached to the positional subspaces themselves, not to a chosen sine/cosine phase convention.

Proof of Theorem 2. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr} \Gamma = \sum_{i=1}^K \text{tr} I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (14). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 3. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (20)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (21)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 , rather than comparing the unwhitened Taylor coefficients. Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (22)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (23)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (24)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1}M) + O(\epsilon^6) = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6), \quad (25)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \rightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (26)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. This is a limit for a declared fixed measure; it does not identify which slow channels a trained model uses. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram. The fixed-range trained-model contrast of §C.1 instead follows its paired empirical protocol.

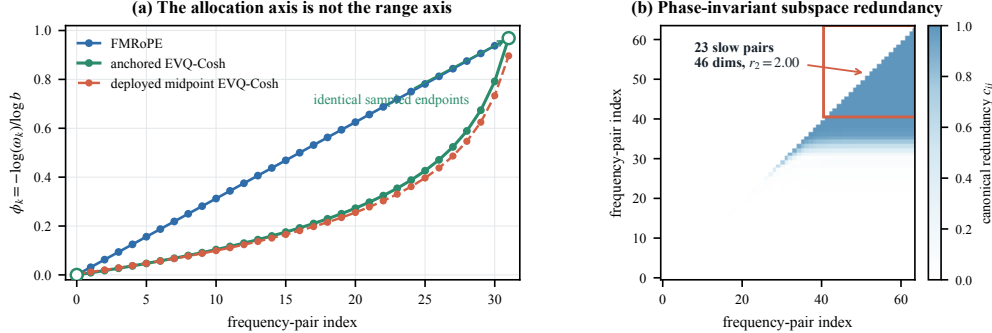


Figure 4: Where a finite frequency table spends its budget. (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

F.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

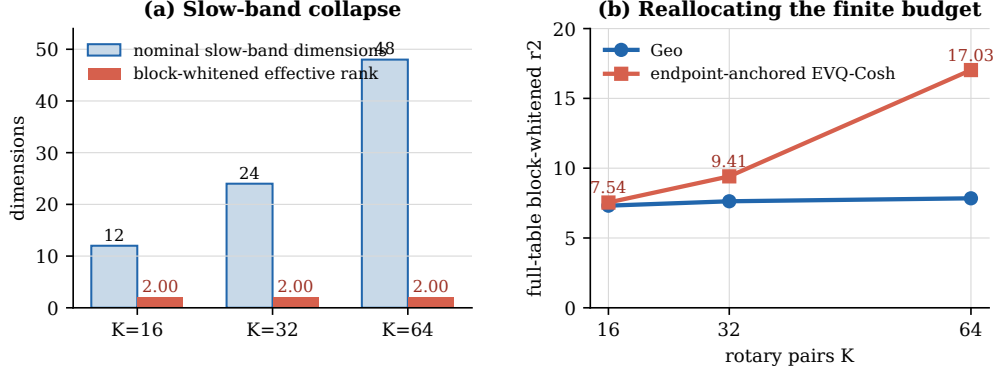


Figure 5: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows. These are static positional-basis diagnostics; the trained effect is identified separately in App. H.

Proposition 7 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (27)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (28)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (29)$$

so every entry of the cross-Gram in (19) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_*((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (28) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k \Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 7 attains the algebraic rank maximum on the declared interval while imposing an exact recurrence just outside it. It does not say that K_{par} is a global capacity over arbitrary frequencies, nor does it survive a change of measure: the causal separation histogram is triangular rather than uniform and does not make the same lattice orthogonal. The exact parity construction and the small-model checks are reproduced by `analysis/full_rope_audit/verify_small_models.py`; the uniform-versus-causal measure comparison is exposed by `scripts/analysis/third_axis_ceiling.py`–`decompose`. These are static basis statements, not LM-quality or extrapolation predictions.

F.3 TWO STATIC ORDERING COUNTEREXAMPLES

Even within a fixed declared measure, a cheaper static proxy need not order tables like the full basis, and an ordering at one interval need not persist at another. The following two finite tables are existence counterexamples; no claim of continuous global optimisation is needed.

Cosine-only collision can choose the lower-rank table. Use $L=4096$, $b=500,000$, $K=64$, continuous uniform $\Delta \in [0, L]$, and pin both physical endpoints to $[2 \times 10^{-6}, 1]$. Let A be the best-found schedule returned by the declared heuristic for cosine-only collision and B the corresponding best-found schedule for full-subspace collision, using their distinct harmonic-augmented candidate pools. The cosine proxy prefers A ,

$$C_{\cos}(A) = 2.62 \times 10^{-9} < 1.44 \times 10^{-8} = C_{\cos}(B), \quad (30)$$

but the phase-invariant block-whitened basis has the opposite order,

$$r_2(A) = 64.48 < 128.00 = r_2(B). \quad (31)$$

Here $C_{\cos}$ is the mean squared normalised cosine-cosine overlap. The counterexample isolates why one phase slice cannot replace the full two-plane object: a small cosine overlap can hide redundancy in the sine and cross terms.

Full-subspace collision can reverse across lengths. Under the same base, endpoints, and training-interval convention, the tracked counterexample search emits two fixed $K=16$ tables A and B . Writing C_M for their mean phase-invariant full-subspace collision when the uniform measure spans $[0, M]$, their ordering is

$$C_L(A) = 0.45837 < 0.61695 = C_L(B), \quad (32)$$

$$C_{2L}(A) = 0.45834 > 0.41132 = C_{2L}(B), \quad (33)$$

$$C_{4L}(A) = 0.45830 > 0.24763 = C_{4L}(B). \quad (34)$$

Recomputing the same schedules on discrete uniform separations $\Delta = 0, \dots, M-1$ preserves the reversal. Thus even the correct phase-invariant collision is conditional on the declared interval; minimising it at L does not supply an extrapolation ordering at $2L$ or $4L$.

Numerical owner and reproduction convention. The digits in Eqs. (31)–(34) are frozen outputs produced by `scripts/analysis/full_rope_collision_audit.py` with `-length 4096 -base 500000 -candidate-grid 1025 -random-schedules 3000`; the counterexample RNG seed is 20260819 and the JSON stores each schedule and its float64 hash. These are best-found points returned by the declared greedy-plus-exchange heuristic in metric-specific candidate pools, not certified pool or continuous optima. The first inequality is also reported with raw full entropy rank $93.44 < 120.03$; the stable-rank ordering above is the quantity governed by Theorem 2.

Together, the parity recurrence and the ordering reversals separate three questions: dimension over a declared interval, multiscale coverage outside it, and behaviour after training. The fixed-support experiments evaluate the last question, while Table 2 shows why learned coefficients must stay with the basis that trained them.

F.4 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 6. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This exact statement covers *fixed, position-independent, invertible* compensation. Approximate re-training on finite data, nonlinear maps, and changes to V/O or the residual path are separate intervention classes.

F.5 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 4 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (35)$$

the main-text collision functional is (15), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (36)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (37)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (38)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (39)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (38) and (39) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (40)$$

which is main-text Theorem 4. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1) = \tau / \sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (40).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (41)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The homogeneous Cosh solution has an analytically invertible CDF, turning the continuous optimum into an explicit finite table. The functional test below checks the resulting allocation against a closed-form cosine-feature slice, and the trained comparisons supply the behavioural test.

F.6 COSINE-SLICE DIAGNOSTIC FOR THE SURROGATE-DERIVED TABLE

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ supplies the closed-form density. We validate its allocation functionally against the closed-form cosine-feature kernel $K_{\text{cos}}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This tests whether the same direction holds for the declared cosine-feature slice.

Table 5 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute K_{cos} at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii}K_{jj})$ and Shannon effective rank (the exponential of the cosine-feature Gram eigenvalue distribution). The inclusive grid $u_k = k/(K-1)$ exposes the complete diagnostic interval. This slice-level static diagnostic is separate from the phase-invariant full-subspace geometry of Theorem 2; trained midpoint tables are evaluated in §E. The 12 reported configurations comprise a six-length text sweep at fixed $b=500\text{K}$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$.

Table 5: Cosine-slice surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases Shannon effective rank of the cosine-feature Gram under the closed-form kernel K_{cos} in all 12 tested configurations. Theorem 2 supplies the separate phase-invariant full-subspace geometry; §E evaluates trained midpoint tables.

Configuration	Collision C		Shannon rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32, b=500\text{K}, \text{sweep } L$</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32, L=2048, \text{sweep } b$</i>				
$b=10\text{K}$	34.8	19.2 (−45%)	13.0	17.3
$b=100\text{K}$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16, L=32, \text{sweep } b$</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1\text{K}$	42.3	28.0 (−34%)	2.8	3.9
$b=10\text{K}$	55.4	39.8 (−28%)	2.2	2.9
$b=50\text{K}$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under K_{cos} in all 12 configurations (24%–92%), and Shannon effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

F.7 SINGLE-CROSSING BUDGET SHIFT

Lemma 8 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (42)$$

Proof of Lemma 8. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving

$\rho_\tau(\phi_c) = 1$ gives (42). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\text{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1 - \phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

F.8 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 9 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (43)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §F). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (44)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (45)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (43). $\square$

An immediate corollary follows.

Corollary 10 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (46)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity checks the two surrogate energy terms exactly; trained-loss evidence is supplied by §E.

F.9 ASSUMPTION-BOUND OPERATING CALCULATION AND FINITE TESTED RECEIPT

The variational theorem selects the Cosh *shape*; it does not select a finite strength for every model or protocol. The implementable reference used in the standard full-RoPE text experiments is

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2}) / K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (47)$$

Here $c=1$ is the pre-training zero-search convention, not a fitted optimum; each protocol records its realised τ , grid, support, and rotary width.

Assumption-bound calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (48)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (49)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1-\phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (50)$$

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives only the leading balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (51)$$

Thus $d_{\text{head}}/\sqrt{L}$ is an assumption-bound scaling factor, while the coefficient remains utility-, base-, and convention-dependent. The reported experiments use the exact Cosh warp, including where small- τ truncation is not accurate; Eq. (51) is not a behavioural selector, a support-invariance statement, or a derivation of $c=1$.

Finite tested receipt. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entryptpoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. These finite points support a fallible zero-search operating prior relative to the tested Geo baseline. They do not establish a continuous basin, basin bounds, interpolation guarantees, near-optimality, or a global optimum.

F.10 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (52)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (53)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$, $m_\tau = \tau / \sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (54)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (55)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoff, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (56)$$

This expansion describes a smooth quantization objective. Its powers of K do not imply that the benefit of EVQ in a trained model increases as rotary width decreases. Anchoring leaves no interior freedom at $K = 2$, when the geometric and EVQ grids coincide. The MLA result at one fixed $K = 16$ also cannot test a budget interaction. The finite-distance bounds in App. A and the registered factorial experiment keep these mathematical and empirical questions separate.

Large- τ regime. At large τ , $m_\tau = \tau / \sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

G EXPERIMENTAL DETAILS

G.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. H and J. This section records the scratch-model and cross-modal experiments that remain supporting evidence. The learned-frequency comparator appears in App. I.

G.2 REPRODUCIBILITY SNAPSHOT

The evidence ladder separates causal identification from architecture, scale, adaptation, frozen-checkpoint, and cross-modal consequences. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 7 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500\text{K}$ unless stated otherwise.

G.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

G.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T} = 2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ ($0.53\times$), which selects the supporting video configuration.

Table 6: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. H.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. H.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. H.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. K.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. G.3/J.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. J.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. J.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. J.1/J.3
Cross-modal scope	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. G.5

Table 7: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 2–4 (750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

G.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 9 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames, extending the scope check to bidirectional 3D RoPE.

Table 8: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

Table 9: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bi-directional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

G.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed mid-point grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 2 separately measures incompatibility after a frozen post-hoc swap.

H IDENTIFICATION PROTOCOL AND PROVENANCE

H.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMROPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R\frac{k}{K-1}\right). \quad (57)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 10: Exact-range anchored EVQ-COSH minus FMROPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean±SD	Seed 42	Seed 137	Seed 256	Mean±SD
256	+0.033	+0.029	+0.017	+0.026 ± 0.008	+0.033	+0.029	+0.017	+0.026 ± 0.008
512	−0.478	−0.271	−0.094	−0.281 ± 0.192	+0.061	+0.053	+0.067	+0.060 ± 0.007
1,024	−0.205	−0.191	−0.132	−0.176 ± 0.039	+0.182	+0.156	+0.344	+0.227 ± 0.102
2,048	−0.113	−0.181	−0.143	−0.146 ± 0.034	+0.279	+0.399	+0.701	+0.460 ± 0.218

Under *fixed*, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under *target-matched*, FMROPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Under the tested target-retargeted policy, the ordering of the two tested policies reverses. The two columns therefore establish that the fixed-support result is conditional on the selected range; they do not identify a support-by-allocation interaction law. The *fixed* column isolates allocation when one table serves multiple lengths (§C.1).

H.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMROPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

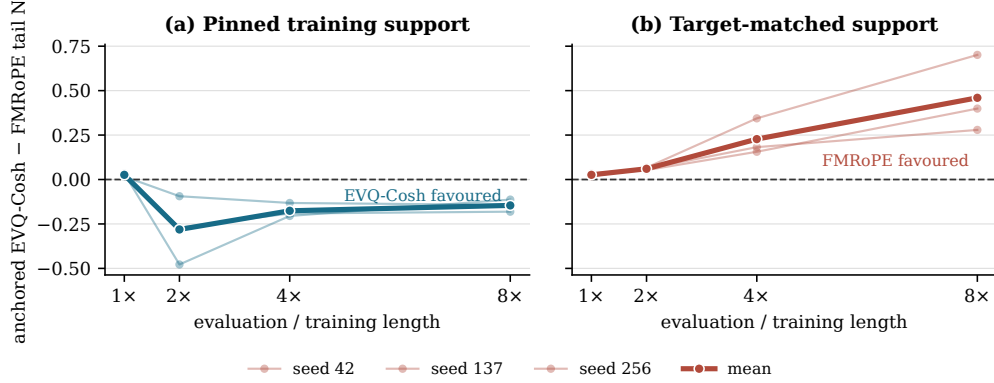


Figure 6: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

Table 11: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. This is a two-seed mechanism replication of weight-table co-adaptation, not a replacement for the three-seed exact-range effect and not evidence that either derived table is globally optimal.

H.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 train stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the 1.25 $\times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at 0.75 $\times$, 1.00 $\times$ (the formula point), 1.25 $\times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Table 12: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times / 4 \times / 8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. H).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. The improving direction points inward at both tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream (Merity et al., 2017), a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §E.4.

I LEARNED-FREQUENCY COMPARATOR

Table 13: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8\text{K}$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §C.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq (100 $\times$ positional LR)	32	183.6	455.3	−11.4%
same, 10 $\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 13 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at 10 $\times$ and 100 $\times$ the base rate, and the stronger 100 $\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §C.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). In this protocol, the learned table improves over Geo but remains weaker than EVQ-COSH; the comparison does not identify whether the gap comes from the final table, optimization dynamics, or the positional-parameter learning rate.

Table 14: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	1 ×	2 ×	4 ×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

J MATURE-SCALE PROTOCOLS

J.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the endpoint grid $u_k=k/K$ with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K *niah_single_1*, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoints are kept separate in the discussion below and in Table 14.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled Q/K *metrics.json* records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 14 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64

at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/ +0.0381/ -0.0437/ -0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

The identity is scientific rather than bitwise trainer pairing. The EVQ-COSH branch uses the reviewed Hugging Face single-GPU loop, whereas the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. Different trainer stacks remain a residual implementation confound, so this is scale persistence from one same-initialisation/same-recipe pair, not a bitwise paired trajectory or a multi-seed causal estimate.

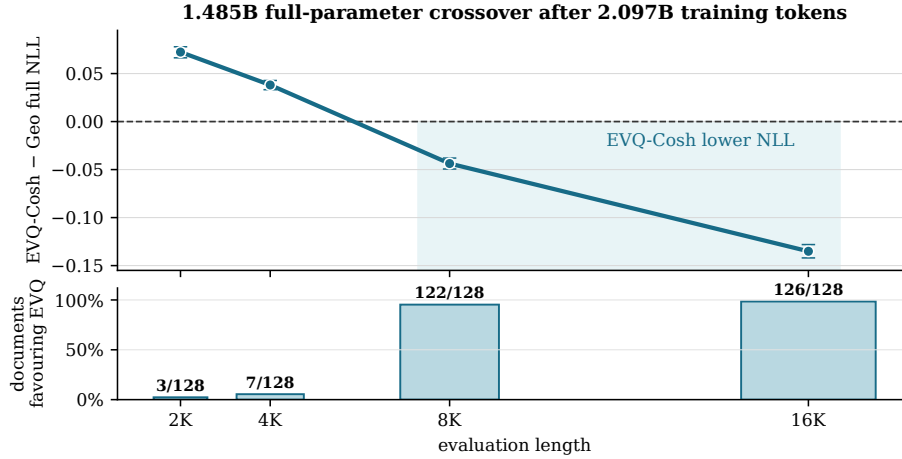


Figure 7: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

J.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Fig. 2 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The two reference rows do not: Native retains the pretrained table, while official Transformers YaRN applies its frequency-dependent transport. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $m_k = (1 - \tilde{U}_k)^2$ in Eq. (18), so less unique pairs move farther.

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalued, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are $[54.88, 64.80]$ points for OLMo and $[0.25, 17.50]$ points for Qwen; the paired

Table 15: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Fig. 2. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 16 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 16: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The complete OLMo deployment policy is consequently a separate bundled intervention: exact Native is selected when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Table 17: Compact zero-training deployment endpoints on the frozen OLMo checkpoint. These are complete-policy comparisons, not pure- z effects.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$, so the result is not universal task dominance. The QA arms use identical rows, and every post-chat input plus generation reserve fits within 16K. Figure 2 remains the allocation-only contrast.

J.3 LLAMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$;

Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. The contrast is therefore the end-to-end effect of assigning different frequency substrates under a matched adaptation pipeline; it does not isolate pure interior shape because the Native endpoint grid and midpoint-Cosh grid also differ in finite-grid embedding. Under this protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry; it changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 14 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

K SUPPORTING RESULTS

K.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified empirical $\tau=1.414$ setting. Because the compressed rotary subspace does not instantiate the full-RoPE $d_{\text{rot}}=d_{\text{head}}$ convention in Eq. (47), this experiment tests frequency allocation at fixed architecture rather than the operating rule.

The tracked aggregate is backed by the exact recovered three-seed evaluator JSON and retains every seedwise row. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree. Original checkpoint and training-data artifacts are not packaged, so this row establishes the reported system ordering, not checkpoint-level reproduction or a theorem fixing τ from d_{rope} .

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

EVQ-COSH reduces 16K PPL by 31.1% ($138.8 \rightarrow 95.6$), while 8K PPL changes by $+0.9\%$. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this architecture and protocol; the multiplier overlay is the run-specific MLA wavelength-blend operator, not the paper’s pure- z control or an exact reproduction of another scaling method.

Table 18: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9